

ALEYSK, Russian Federation

WMO: 299370

Lat: 52.517N

Lon: 82.767E

Elev: 176

StdP: 99.23

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.3	-31.2	-37.4	0.1	-34.2	-34.3	0.2	-31.0	17.0	-6.0	15.2	-7.1	0.9	200	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	31.4	19.0	29.6	18.4	27.9	17.8	20.9	27.9	20.0	26.7	19.1	25.7	2.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.8	24.1	17.6	12.9	22.9	16.6	12.1	22.1	61.5	27.9	58.2	26.6	55.3	25.8	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	10.3	8.3	DB	-36.1	34.6	3.9	2.2	-38.9	36.2	-41.2	37.5	-43.4	38.7	-46.3	40.4	(4)
(5)			WB	-36.2	23.1	3.9	1.9	-39.0	24.4	-41.3	25.6	-43.5	26.6	-46.3	28.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-16.6	-13.8	-5.5	5.7	13.2	19.1	20.4	18.1	11.7	4.6	-5.4	-12.8	(6)
(7)		DBStd	14.33	8.34	7.64	6.57	5.73	5.10	3.89	3.23	3.68	4.33	5.14	7.88	7.67	(7)
(8)		HDD10.0	3524	823	668	480	149	27	1	0	1	31	177	462	706	(8)
(9)		HDD18.3	5677	1082	901	738	378	170	38	17	49	202	425	711	964	(9)
(10)		CDD10.0	1088	0	0	0	22	127	273	321	252	82	11	0	0	(10)
(11)		CDD18.3	198	0	0	0	0	12	61	79	42	3	0	0	0	(11)
(12)		CDH23.3	2093	0	0	0	14	207	592	702	496	80	3	0	0	(12)
(13)		CDH26.7	622	0	0	0	2	56	175	217	152	19	0	0	0	(13)
(14)	Wind	WSAvg	2.7	3.0	2.9	2.9	3.0	2.8	2.3	1.8	1.9	2.2	3.0	3.4	3.7	(14)
(15)	Precipitation	PrecAvg	456	26	20	25	29	40	54	68	44	33	37	43	36	(15)
(16)		PrecMax	634	52	33	63	72	96	157	139	128	92	108	82	74	(16)
(17)		PrecMin	281	4	1	2	3	12	16	16	4	6	9	14	10	(17)
(18)		PrecStd	96	13	9	16	17	22	33	30	25	20	22	18	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	2.6	11.1	25.3	31.5	32.7	33.5	33.9	29.6	22.7	12.7	3.0	(19)
(20)			MCWB	0.6	0.5	6.7	13.0	16.6	18.7	20.8	20.1	16.2	11.5	7.0	0.9	(20)
(21)		2%	DB	-0.2	0.5	7.2	21.3	28.1	30.6	31.1	30.3	25.9	18.5	9.1	1.1	(21)
(22)			MCWB	-1.3	-1.0	3.9	11.4	15.3	18.2	20.2	19.0	15.0	10.1	5.4	-0.3	(22)
(23)		5%	DB	-2.5	-1.3	4.6	18.4	25.7	28.6	29.2	28.1	22.9	15.5	6.2	-0.7	(23)
(24)			MCWB	-3.7	-2.7	2.1	10.2	14.1	17.8	19.3	18.0	13.9	8.9	3.5	-1.8	(24)
(25)		10%	DB	-4.8	-3.3	2.8	15.2	23.0	26.8	27.4	26.1	20.1	12.4	3.6	-2.4	(25)
(26)			MCWB	-5.8	-4.5	0.9	8.6	13.4	17.5	18.9	17.4	12.5	7.5	1.8	-3.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	1.0	6.8	14.0	18.4	21.1	22.9	21.4	17.8	12.3	8.1	1.2	(27)
(28)			MCDWB	2.1	2.2	10.8	22.9	28.4	28.1	30.1	29.7	26.4	19.1	10.7	2.4	(28)
(29)		2%	WB	-1.3	-0.9	4.0	12.1	16.4	19.8	21.5	20.1	15.7	10.7	5.5	-0.1	(29)
(30)			MCDWB	-0.3	0.5	6.7	19.6	25.2	26.9	28.0	27.8	23.6	17.0	8.8	1.0	(30)
(31)		5%	WB	-3.7	-2.7	2.3	10.6	15.3	19.0	20.5	19.0	14.5	9.4	3.6	-1.8	(31)
(32)			MCDWB	-2.7	-1.4	4.4	17.2	23.1	26.2	26.9	26.3	21.6	15.0	6.1	-0.8	(32)
(33)		10%	WB	-5.8	-4.4	1.0	9.1	14.1	18.2	19.7	18.0	13.2	7.9	1.8	-3.3	(33)
(34)			MCDWB	-4.8	-3.2	2.6	15.1	21.4	25.0	25.7	24.8	19.5	12.0	3.6	-2.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	9.4	9.1	10.9	12.3	11.2	10.6	12.2	12.4	9.8	7.1	7.9	(35)
(36)			MCDBR	9.0	9.6	9.9	16.6	16.3	14.3	13.5	15.8	17.6	16.2	9.1	9.2	(36)
(37)			MCWBR	8.3	8.8	7.3	8.5	6.8	5.4	5.1	6.1	8.0	8.7	6.3	8.5	(37)
(38)		5% WB	MCDWR	9.1	9.7	9.4	14.7	13.5	11.9	11.5	13.8	15.7	14.8	8.2	9.0	(38)
(39)			MCWBR	8.5	9.0	7.2	8.3	6.5	5.1	5.1	6.0	7.6	8.5	6.1	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.338	0.391	0.419	0.418	0.413	0.437	0.412	0.374	0.361	0.347	0.310	(40)	
(41)		taud	2.014	1.944	1.898	2.096	2.168	2.264	2.197	2.241	2.290	2.210	2.039	1.962	(41)	
(42)		Ebn at Noon	639	727	767	809	833	841	810	810	797	712	571	534	(42)	
(43)		Edh at Noon	81	118	152	142	139	128	135	122	103	90	79	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.19	2.26	3.70	4.71	5.77	6.36	5.74	4.99	3.62	1.93	1.02	0.83	(44)	
(45)		RadStd	0.09	0.14	0.37	0.33	0.49	0.43	0.37	0.30	0.31	0.24	0.13	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.65	-0.52	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	-0.53	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page